

Do as Sonographers Think: Contrast-Enhanced Ultrasound for Thyroid Nodules Diagnosis via Microvascular Infiltrative Awareness

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Abstract—Dynamic contrast-enhanced ultrasound (CEUS) imaging can reflect the microvascular distribution and blood flow perfusion, thereby holding clinical significance in distinguishing between malignant and benign thyroid nodules. Notably, CEUS offers a meticulous visualization of the microvascular distribution surrounding the nodule, leading to an apparent increase in tumor size compared to gray-scale ultrasound (US). In the dual-image obtained, the lesion size enlarged from gray-scale US to CEUS, as the microvascular appeared to be continuously infiltrating the surrounding tissue. Although the infiltrative dilatation of microvasculature remains ambiguous, sonographers believe it may promote the diagnosis of thyroid nodules. We propose a deep learning model designed to emulate the diagnostic reasoning process employed by sonographers. This model integrates the observation of microvascular infiltration on dynamic CEUS, leveraging the additional insights provided by gray-scale US for enhanced diagnostic support. Specifically, temporal projection attention is implemented on time dimension of dynamic CEUS to represent the microvascular perfusion. Additionally, we employ a group of confidence maps with flexible Sigmoid Alpha Functions to aware and describe the infiltrative dilatation process. Moreover, a self-adaptive integration mechanism is introduced to

dynamically integrate the assisted gray-scale US and the confidence maps of CEUS for individual patients, ensuring a trustworthy diagnosis of thyroid nodules. In this retrospective study, we collected a thyroid nodule dataset of 282 CEUS videos. The method achieves a superior diagnostic accuracy and sensitivity of 89.52% and 94.75%, respectively. These results suggest that imitating the diagnostic thinking of sonographers, encompassing dynamic microvascular perfusion and infiltrative expansion, proves beneficial for CEUS-based thyroid nodule diagnosis.

Index Terms—Contrast-enhanced ultrasound, thyroid nodules, dynamic perfusion, infiltrative expansion.

I. INTRODUCTION

THYROID nodules are the most prevalent thyroid disorder and can adversely impact human health. The thyroid, being an organ with a well-developed microvascular system, exhibits distinct vascular structures in both benign and malignant thyroid nodules. Thyroid ultrasound (US) is the preferred imaging technique for thyroid examination, owing to its numerous advantages, such as portability, affordability, real-time imaging capabilities, and non-invasiveness.

A. Gray-Scale Ultrasound Image Analysis for Thyroid Nodule Segmentation and Classification

In recent years, there has been a growing utilization of computer-aided analysis methods for detecting, segmenting, and classifying thyroid nodules using traditional grayscale ultrasound (US). Notably, researchers increasingly incorporate domain knowledge to enhance performance. Liu et al. [1] utilized the false foreground (FFG) region surrounding the nodule as a prior spatial constraint to improve sensitivity to small nodules. Chen et al. [2] introduced a novel semi-supervised boundary refinement shadow perception network (SABR-Net), incorporating prior knowledge of shadow regions for more precise thyroid nodule segmentation. Sun et al. [3] proposed a dual-path convolutional neural network, featuring a region path and a shape path, and inserted a soft shape supervisor block between the two paths to enhance thyroid nodule segmentation. Gong et al. [4] proposed the Thyroid Region Priori Guided Feature Enhancement Network (TRFE+), which utilizes an adaptive gland region feature enhancement module to learn the nodule size, gland location, and nodule position simultaneously. Furthermore, Zhao et al. [5] integrated

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tissue anatomy domain knowledge into their model with an attention mechanism for benign-malignant thyroid nodule classification. Lu et al. [6] achieved significant diagnostic performance in distinguishing benign and malignant thyroid nodules by employing a deformable attention network guided by manually labeled nodule masks. Shi et al. [7] proposed a knowledge-guided synthetic medical image adversarial augmentation method for thyroid US image classification. The methods described above primarily utilized a single static US image frame for diagnosing thyroid nodules. To fully leverage the characteristics of thyroid US videos, Wu et al. [8] introduced a post-processing method called CacheTrack-YOLO. This method capitalizes on the contextual relationships between US video frames, showing promising performance in the analysis of thyroid US videos. However, it is important to note that CacheTrack-YOLO exclusively employs a single grayscale US modality, overlooking other US modalities that may offer significant diagnostic value.

B. Diagnostic Value of Dynamic CEUS

Contrast-enhanced ultrasound (CEUS) is a modality of functional imaging that can assess the intensity of vascular perfusion and hemodynamics in the thyroid nodule [9]; thus, it is considered a valuable new approach in the determination of benign vs. malignant nodules. Currently, in practical clinical applications, sonographers are able to observe real-time, dynamic information regarding microvascular perfusion of tissues by using the intravenous injection of ultrasound contrast agent (UCA) in conjunction with conventional grayscale US, as illustrated in Fig.1. Specifically, dynamic CEUS can be acquired simultaneously with conventional grayscale US. The combination of dynamic CEUS with grayscale US has been increasingly utilized for the diagnosis of thyroid nodules and is becoming a commonly employed screening modality [10]. Zhao et al. demonstrated that the combination of multimodal ultrasound, including dynamic CEUS and grayscale ultrasound, with the Thyroid Imaging Reporting and Data System (TI-RADS), significantly enhances the accuracy of distinguishing between benign and malignant thyroid nodules [11]. The enhancement pattern of lesions identified through grey-scale US -assisted contrast-enhanced ultrasound (CEUS) is widely regarded as the most reliable predictor for diagnosing malignant thyroid nodules [12]. Notably, the dual-image display format, which combines CEUS and grayscale US, is progressively being employed across a diverse range of organs [12], [13].

In recent years, computer-aided diagnostic methods have been increasingly employed for diagnosing thyroid nodules using dynamic CEUS. Traditional CEUS analysis methods [14] utilize the time intensity curve (TIC) of the lesion area to obtain functional parameters (*e.g.*, nodule arrival time, time to peak, *etc.*) for computer-aided diagnosis. While these functional parameters provide valuable insights into the perfusion characteristics observed in CEUS videos, they often overlook the rich morphological features. Presently, there is a growing focus on dynamic CEUS imaging, which captures perfusion information related to the lesion, with the potential to enhance the precision of lesion diagnosis.

C. Dynamic CEUS Analysis

1) *Dynamic CEUS Analysis Based on Deep Learning*: With the advancement of deep learning, CEUS-based lesion diagnosis methods primarily rely on convolution neural networks (CNN) to extract spatial-temporal features from dynamic CEUS for disease diagnosis. For instance, Qin et al. [15] employed the ResNet50 and VGG16 classification networks to extract features from CEUS images. Yang et al. [16] highlighted the significance of the time sequence of CEUS and combined CNN with a Temporal Sequence Regression Mechanism (TSRM) to extract dynamic enhanced features of the lesion area for automatic diagnosis. Existing deep learning approaches mainly use CNNs for dynamic CEUS analysis, however, the restricted receptive field of CNNs poses a challenge in accurately simulating Time- Intensity Curves (TIC), which are crucial for clinical diagnoses in CEUS. This limitation hinders the accurate modeling of CEUS time series. Although Transformers have shown remarkable ability in modeling time series with extensive dependencies [S9], their direct integration into the automated analysis of CEUS faces challenges due to the lack of publicly available datasets. Therefore, we propose a CNN-based architecture that includes a Transformer-like module called Temporal Projection Attention to effectively capture the characteristics of dynamic CEUS time series.

Nevertheless, the precise identification of the lesion area, which is essential for disease diagnosis, still necessitates time-consuming manual labeling. To solve this issue, Chen et al. [17] proposed a joint segmentation and classification network that improves diagnostic accuracy by precisely localizing lesion locations without pixel-level manual delineation. Gan et al. introduced a multi-task algorithm that utilizes complementary information from segmentation and classification to prioritize features within the lesion region, aiming for precise carotid plaque diagnosis [18]. Xie et al. [19] proposed a mutual guidance network for segmentation and classification, where the mask generated by the segmentation task is utilized to guide the classification task and achieve precise lesion localization. Although these algorithms guide the classification task using binary masks obtained from the segmentation module, the use of binary masks disregards the uncertainty of lesion boundaries [20] and the diagnostic value of the surrounding areas of the tumor. In other words, when training the network to prioritize the features of the lesion area and eliminate irrelevant information, we should not completely disregard the features of the peri-tumor area that radiologists focus on during the diagnostic process [21], as well as the influence of uncertain lesion boundaries [22]. Inspired by these studies, our study aims to jointly achieve lesion area identification and differential diagnosis, aligning with the perspective of sonographers.

2) *Dynamic CEUS Analysis Enhanced by Clinical Information*: There are many microvasculatures around the tumor, which appeared to be continuously infiltrating the surrounding tissues [23]. CEUS can reflect the invasion of surrounding blood vessels by lesions. Some clinical studies [24] have found that the measurement discrepancies in lesions sizes, especially larger measurement in dynamic CEUS compared to

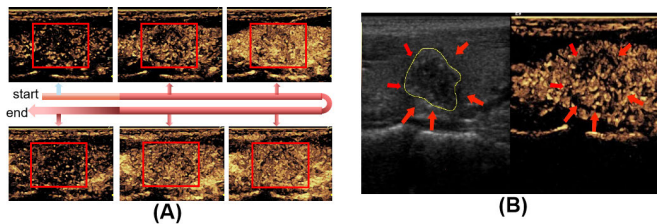


Fig. 1. (A) The dynamic frames in CEUS video. In practice, sonographers analyze the lesion by observing the microvascular perfusion in the lesion area (red box). From the start to the end of the timeline, the three colors represent the change in intensity of CEUS video over three different time periods. (B) The size of thyroid nodules on CEUS is significantly larger than detected on gray-scale US because of the phenomenon of microvascular infiltrative dilatation. The yellow line indicates lesion area on gray-scale US, while the red arrow corresponds to the infiltration area or continuous lesion expansion on CEUS.

gray-scale US, are more likely to be observed in malignant tumors rather than benign lesions. Fig.1.(B) illustrates this phenomenon, showcasing larger lesion sizes in dynamic CEUS compared to gray-scale US due to microvascular invasive expansion. The microvascular infiltration described here is distinct from Microvascular Invasion (MVI) [25] observed in hepatocellular carcinoma. *While sonographers may not have a clearly defined understanding of the microvascular invasive expansion, they believe that it could enhance the diagnosis of thyroid nodules [26].* Despite this, the impact of microvascular infiltrative expansion on CEUS analysis remains unexplored. Particularly, the question of whether awareness of infiltrative area can enhance diagnostic accuracy has not been explored. This prompts us to further explore the impact of this specific infiltration-aware mechanism on the diagnosis of thyroid nodules.

Currently, some researchers have also suggested the use of gray-scale US as a complementary tool for disease diagnosis alongside CEUS. By combining gray-scale US modality, Wan et al. [27] proposed a hierarchical temporal attention network, leveraging spatial feature enhancement for disease diagnosis based on dynamic CEUS. Chen et al. [28] proposed a domain-knowledge-guided temporal attention module for breast cancer diagnosis. Furthermore, Chen et al. [29] proposed the cross-modal interactive learning by introducing optical flow information and gray-scale US. These approaches entail straightforwardly concatenating features derived from both CEUS and gray-scale US to create a unified representation, thereby enhancing the diagnostic process. However, they overlook the inherent distinctive characteristics of each US modality and do not consider the varying diagnostic value for individual patients [30].

D. Our Contribution

Guided by the US characteristics [31] and diagnostic guidelines for thyroid nodules, it is acknowledged that different imaging features contribute unequally to the diagnostic task and offer varying levels of information. On the one hand, during the diagnostic process, sonographers tend to pay more attention to CEUS video frames within the peak period, aligned with the fitted TIC. On the other hand, due to the inherent characteristics of US imaging, certain modalities

might not provide useful diagnostic information for specific patients, resulting in unequal levels of information across modalities for a particular diagnostic task [32]. Unfortunately, current CEUS models assisted by gray-scale US often overlook the impact of differing amounts of information provided by various features or modalities in a specific task. In order to align our model with the thought process of sonographers, we introduce a dynamic informativeness integration mechanism that enables the model to comprehensively consider the impact of informativeness differences in diagnosing thyroid nodules among different patients. To sum up, during the CEUS based thyroid nodules diagnosis, sonographers typically focus on the following points:

1) *Dynamic microvascular perfusion.* As shown in Fig.1 (A), the clinically acquired CEUS video captures dynamic changes in relative intensity (microvascular perfusion pattern) throughout the entire examination.

2) *Infiltrative expansion of microvasculature.* Microvascular around nodules consistently infiltrates the surrounding tissue. While sonographers may not have a clearly defined understanding of the microvascular invasive expansion, they believe that it could enhance the diagnosis of thyroid nodules.

3) *Self-adaptive information integration.* The sonographers consider different informativeness of CEUS and gray-scale US for different patient during the diagnosis. Thus, conducting a self-adaptive integration mechanism tailored to individual patients is deemed beneficial for enhancing the trustworthiness and explainability of the model.

In this paper, we develop a deep model designed to emulate the diagnostic reasoning process employed by sonographers. Our model consists of three novel components: 1) Temporal Projection Attention (TPA) is proposed to complement and interact with the semantic information of microvascular perfusion. TPA operates in the time dimension of CEUS; 2) instead of binary masks, we adopt a group of confidence maps to perceive the infiltrative expansion area from gray-scale US to CEUS of microvasculature, enhancing diagnostic precision; 3) for specific patients, the gray-scale US is adaptively utilized to assist CEUS in completing diagnosis by evaluating the informational contribution of different modalities. Our preliminary work establishing a model to consider the microvascular infiltration for thyroid nodules diagnosis was published in MICCAI 2023 [33]. In this substantially expanded journal paper, we offer new contributions in the following aspects: 1) a self-adaptive integration mechanism is proposed to dynamically utilize gray-scale US to assist CEUS for different patients, resulting in improved diagnosis of thyroid nodules compared to our preliminary work; 2) In-depth validation experiments are implemented to validate the effectiveness and explainability of the proposed framework.

II. METHOD

A. Overview

During the US screening, an US probe is used to scan the lesion, and the dynamic gray-scale US and CEUS images are simultaneously acquired using the dual-image display format.

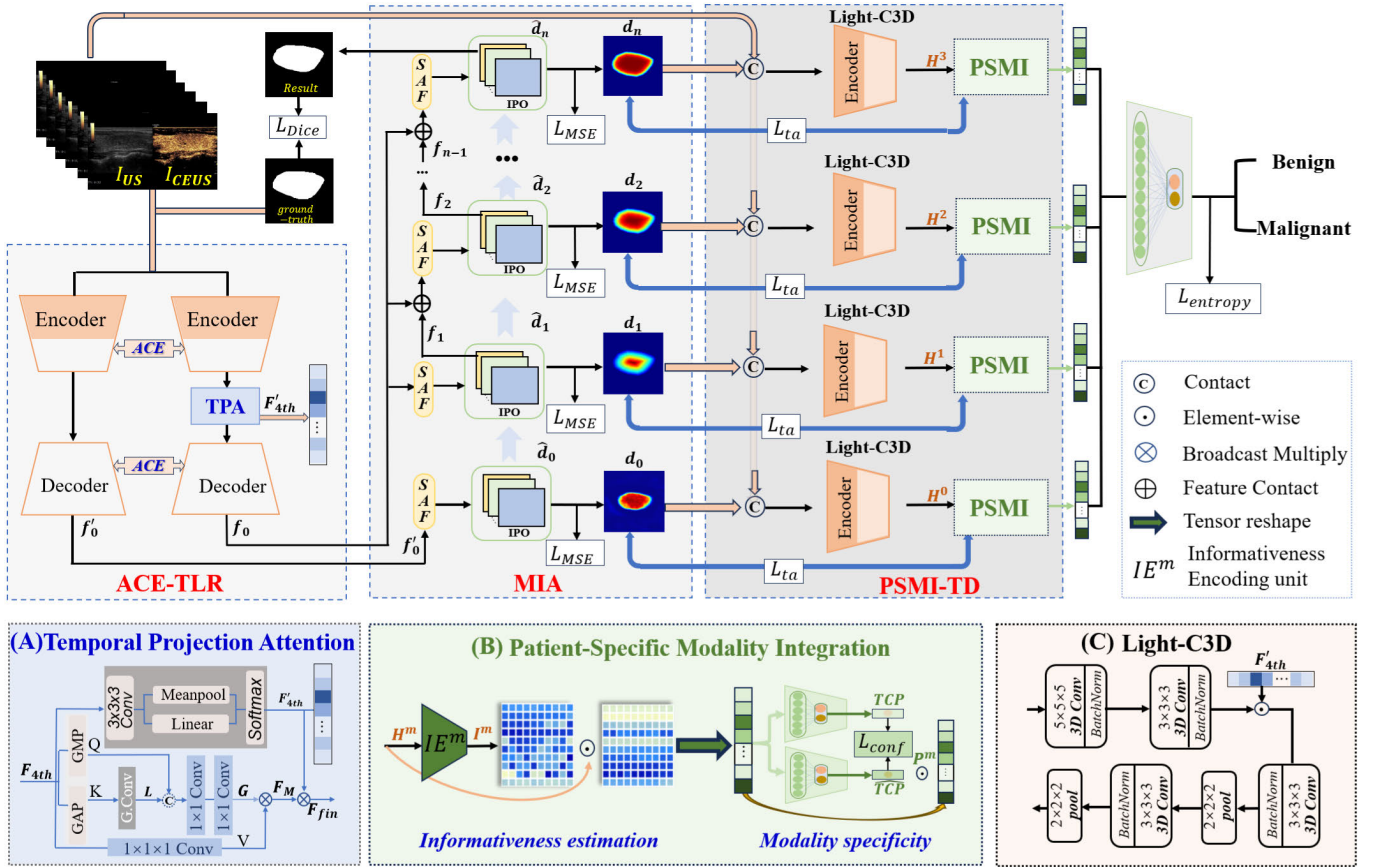


Fig. 2. The architecture of our proposed framework mainly contains three parts: 1) the **Adaptive Channels Exchanging and Temporal-based Lesion Recognition (ACE-TLR)** part including the Adaptive Channels Exchanging (ACE) and Temporal Projection Attention (TPA) modules; 2) the **Microvascular Infiltration Awareness (MIA)** part; and 3) the **Patient-Specific Modality Integration for Thyroid nodules Diagnosis (PSMTD)** part. Moreover, we use three additional subgraphs to represent our three important modules: (A) Temporal Projection Attention module (TPA), (B) Patient-Specific Modality Integration module (PSMI) which involves the estimation of informativeness and performing multimodal classification based on modality specificity; and (C) Light-C3D based classification module.

This allows sonographers to complete the clinical diagnosis by comparing these two modalities. Notably, sonographers have noted that dynamic CEUS can unveil information about the blood supply in tumor microvasculature, resulting in dynamic CEUS indicating larger tumor sizes than those observed with grayscale US. To our knowledge, no investigator has performed CEUS analysis while considering this interesting clinical phenomenon. In addition, current studies on CEUS-based thyroid nodules diagnosis assisted by gray-scale US does not sufficiently consider the specificity of individual patients. As shown in Fig. 2, the architecture of our proposed framework mainly contains three parts. Firstly, the part of Adaptive Channels Exchanging and Temporal-based Lesion Recognition (ACE-TLR) integrates two parts modules including the Adaptive Channels Exchanging (ACE) and Temporal Projection Attention (TPA). The ACE-TLR allows for the automatic recognition of the lesion area from dynamic CEUS, with the assistance of gray-scale US. Secondly, inspired by the observation that tumor lesions and their margins depicted by CEUS may appear larger due to continuous infiltrative expansion compared to those depicted by gray-scale US, we design the Microvascular Infiltration Awareness (MIA) part. This part is designed to learn the process of infiltrative expansion

from gray-scale US to CEUS. Thirdly, the Patient-Specific Modality Integration for Thyroid nodules Diagnosis (PSMTD) is proposed to dynamically combine gray-scale US and the microvascular infiltration maps of CEUS for different patients, ensuring a trustworthy and patient-specific completion of thyroid nodule diagnosis.

The gray-scale US $I_{US} \in \mathbb{R}^{C \times T \times H \times W}$ and CEUS $I_{CEUS} \in \mathbb{R}^{C \times T \times H \times W}$ are fed into the ACE-TLR part, which incorporates an enhanced V-Net [34] with the ACE and TPA modules to identify the relatively distinct lesion areas visible on both gray-scale US and CEUS. Dynamic CEUS provides information about the tumor's blood supply, resulting in larger tumor size and margins depicted in the CEUS compared to gray-scale US. To explore the potential contribution of this interesting clinical phenomenon contributes to the diagnostic task, our MIA module utilizes a set of confidence maps generated through the Sigmoid Alpha Function (SAF) instead of simple binary masks. These confidence maps help detect the infiltrative expansion area of the microvasculature from gray-scale US to CEUS. Subsequently, these confidence maps are fused with I_{CEUS} and I_{US} , and fed into the lightweight C3D-based [35] PSMTD component. This component predicts the probability score of benign or malignant tumors

by considering the varied levels of information provided by different modalities for each specific patient. For the tasks of lesion area recognition and differential diagnosis, the low-level features are interactively shared through the first two layers of encoders in the lesion area segmentation and differential diagnosis tasks. In particular, in the input diagnosis network, each CEUS frame is weighted using the output F'_{4th} of TPA module. This weighting allows the diagnostic model to fit the time intensity curve (TIC) of dynamic CEUS without additional constraints.

B. Adaptive Channel Exchanging for Temporal-Based Lesions Recognition (ACE-TLR)

Accurately identifying the location of lesions can be challenging when relying solely on CEUS modality or an isolated frame of dynamic CEUS captured at different perfusion intervals. In clinical practice, sonographers typically estimate the approximate location of the lesion with the assistance of gray-scale US. Subsequently, they analyze the entire CEUS video to ascertain the tumor boundary with greater precision.

Commonly, the automated recognition of the lesion area using dynamic CEUS presents a substantial challenge, primarily due to the limited semantic information available from only one of the three perfusion phases (*Arterial Phase*, AP; *Portal Venous Phase*, PV-P; *Delayed Phase*, DP). In particular, the semantic information of lesions in the AP and DP cannot be fully represented in a single CEUS image frame. To address these challenges, we propose the TPA module, designed to fuse the semantic information from the entire perfusion period. Our model adopts the V-Net architecture, consisting of four encoder/decoder blocks, with the TPA integrated into the bottleneck. Here we only perform downsampling operations at the spatial resolution of dynamic CEUS. In the temporal dimension, we fuse frame-level information interactively solely through the TPA module. Employing an adaptive and parameter-free ACE strategy facilitates the integration of complementary information between dynamic CEUS and gray-scale US. This strategy allows us to leverage the assistance from gray-scale US. In summary, ACE-TLR part contains two core components: ACE and TPA. ACE involves the adaptive exchange of channels between subnetworks of gray-scale US and CEUS modalities in an end-to-end manner. In addition, TPA is introduced to complement and interact with the semantic details of microvascular perfusion derived from the temporal dimension of CEUS.

1) Adaptive Channel Exchanging (ACE): The core operation of ACE involves the adaptive exchanging of channels between subnetworks of gray-scale US and CEUS modalities in an end-to-end manner. This adaptive exchanging of channels is guided by a single channel importance [36], which is measured by the magnitude of the Batch Normalization (BN) scale factor γ_c^z during training. The training paradigm of ACE aims to minimize the loss of each subnetwork by integrating complementary information from gray-scale US and CEUS modalities through channel exchange. To effectively integrate information between different modalities without introducing

additional training parameters, the process of ACE can be described as follows:

$$\min_{Fu^{1:Z}} \sum_{z=1}^Z \left(Fu^z(x^{1:Z}), y^z \right) + \lambda \|\hat{\gamma}^z\|_1 \quad (1)$$

where $Fu^z(\cdot)$ represents for the fusion of complementary information between different modalities through channel exchange from other modalities. Suppose we have Z modalities and $z \in \{1, 2\}$ represents gray-scale US and dynamic CEUS, respectively. $\hat{\gamma}^z$ represents the scaling factor applied to all channels of BN operation, and we will penalize the ℓ_1 paradigm for a certain fraction $\hat{\gamma}^z$ to ensure sparsity.

BN layers are extensively employed in deep learning to mitigate covariate bias and enhance generalization. As illustrated in Equation 2, particularly when $\gamma_c^z > \theta$, if γ_c^z approaches 0, it implies that x_c^z will contribute very little, if any, valid information to the final prediction task. As a result, it becomes redundant in the training process. Here, to alleviate the negligible impact of a small channel scaling factor γ^z on the final prediction task, we employ channel information from distinct sub-networks to replace channels with a scaling factor below a defined threshold θ . In other words, if the scale factor of the feature map is below the threshold, the feature map will be replaced with feature maps from other modalities that are located at the same position, as described in equation (2):

$$x_c^{z'} = \begin{cases} \gamma_c^z \frac{x_c^z - \mu_c^z}{\sqrt{(\sigma_c^z)^2 + \varepsilon}} + \beta_c^z, & \gamma_c^z > \theta \\ \frac{1}{Z-1} \sum_{z' \neq z} \gamma_c^{z'} \frac{x_c^{z'} - \mu_c^{z'}}{\sqrt{(\sigma_c^{z'})^2 + \varepsilon}} + \beta_c^{z'}, & \gamma_c^z \leq \theta \end{cases} \quad (2)$$

The BN layer normalizes x^z when the scaling factor is greater than θ . Here, x^z and x_c^z represent the feature map of the z -th modality and the c -th channel of the z -th modality, respectively. $z \in \{1, 2\}$ represent gray-scale US and dynamic CEUS; μ_c^z and σ_c^z are the mean and the standard deviation; ε is a number to avoid dividing by zero, and β_c^z is a compensation factor. Conversely, if one channel of a modality has minimal effect on the final prediction (below θ), it is replaced with the average of the other modalities.

2) Temporal Projection Attention (TPA): As shown in Fig. 2, given a feature $F_{4th} \in \mathbb{R}^{C \times T \times \frac{H}{16} \times \frac{W}{16}}$ after four down-sampling operations in the ACE-based encoder, its original 3D feature map is projected to a 2D plane [37] to get keys and queries, denoted as $K, Q \in \mathbb{R}^{C \times \frac{H}{16} \times \frac{W}{16}}$, using global average pooling (GAP) and global maximum pooling (GMP) as temporal projection operations. Here, $V \in \mathbb{R}^{C \times T \times \frac{H}{16} \times \frac{W}{16}}$ is obtained through a single $1 \times 1 \times 1$ convolution. This operation can also filter out the irrelevant background information and highlights the key information [38] of the lesions. Following the temporal projection, a group convolution (*Gonv*) with a group size of 4 is employed on K to extract the local temporal attention $L \in \mathbb{R}^{C \times \frac{H}{16} \times \frac{W}{16}}$. Subsequently, we concatenate L with Q to further obtain the global attention $G \in \mathbb{R}^{C \times 1 \times \frac{H}{16} \times \frac{W}{16}}$ by two

consecutive 1×1 convolutions and dimension expand. Those operations are described as follows:

$$\begin{aligned} K &= Q = GAP(F_{4th}) + GMP(F_{4th}) \\ G &= Expand(Conv(\phi(Conv(\phi(Gonv(K)) \oplus Q)))) \quad (3) \end{aligned}$$

where ϕ denotes the normalization, and “ $\oplus$ ” is concatenation operation. Then, based on the 2D attention G, we perform a broadcast multiplication of V and G to calculate the global temporal attention map $F_M \in \mathbb{R}^{C \times T \times \frac{H}{16} \times \frac{W}{16}}$, thereby enhancing feature representation.

Based on the common clinical experience, each frame of CEUS provides an unequal amount of information for the diagnostic task. The informativeness provided by individual frames at different times roughly aligns with the Time-Intensity Curve (TIC). Therefore, to make the exacted frame-level features information approximating TIC curve, we use parallel average pooling and full connection operation to reweight the frame-level information F_{4th} to obtain the reweighted feature F'_{4th} . The obtained global temporal fusion attention maps F_M are fused with the reweighted feature F'_{4th} to get the output feature F_{fin} . Finally, we input F_{fin} into the decoder of the ACE-TLR to acquire the feature map of the lesion.

C. Microvascular Infiltration Awareness (MIA)

There has been no investigation into the potential influence of microvascular infiltrative expansion on CEUS analysis. More specifically, the impact of incorporating infiltrative area into diagnostic accuracy remains unexplored. Motivated by this research gap, we aim to conduct a study to investigate the effects of an infiltration-aware mechanism on the diagnosis of thyroid nodules. Inspired by the observation that tumors and margins visualized by dynamic CEUS may exhibit larger size than those depicted by gray-scale US due to continuous infiltrative expansion, we design a MIA module to comprehend the process of infiltrative expansion from gray-scale US to dynamic CEUS. We use a series of flexible Sigmoid Alpha Functions (SAF) to simulate the infiltrative expansion of microvasculature by establishing the distance maps [39] from each pixel to the lesion boundary. Here, the distance maps are denoted as the initial probability distribution D_{IN} . Then, we utilize Iterative Probabilistic Optimization (IPO) unit to produce a set of optimized probability distribution maps $\hat{D} = \{\hat{d}_1, \hat{d}_2, \dots, \hat{d}_n\}$ to enhance the awareness of microvascular infiltration in thyroid nodules diagnosis. Based on SAF and IPO, CEUS-based diagnosis of thyroid nodules can effectively leverage the ambiguous information caused by microvascular infiltrative expansion.

1) **Sigmoid Alpha Function (SAF)**: It is generally believed that the differentiation between benign and malignant thyroid nodules is related to the pixels around the boundaries of the lesion [40]. Therefore, we initially construct the initial probability distribution based on the distance between the pixels and the annotation boundaries by using SAF in order to aware the infiltrative areas. Here, SAF can transform a distance

map into a probability map, defined as follows:

$$SAF_{(i,j)} = C * \left(\frac{2}{\frac{-\alpha S_{(i,j)}}{1 + e^{\max(S_{(i,j)})}} - 1} \right) \quad (4)$$

$$C = (1 + e^{-\alpha}) / (1 - e^{-\alpha}); \alpha \rightarrow (0, +\infty) \quad (5)$$

where α is the conversion factor for generating the initial probability distribution D_{IN} (when $\alpha \rightarrow \infty$, D_{IN} is binary mask); C is used to control the function value within $[0, 1]$; (i, j) is the coordinate point in feature map; $S_{(i,j)}$ indicates the shortest distance from (i, j) to lesion's boundaries.

2) **Iterative Probabilistic Optimization (IPO) Unit**: Thyroid nodules display significant variability in size and aspect ratio, often presenting in strip-like formations [41]. Based on the fact that Pixel-Anchor [42] has experimentally demonstrated that large-scale asymmetric convolution can effectively solve the above problem, we use asymmetric convolution in the IPO unit. Asymmetric convolution-based IPO unit can optimize the initial distribution D_{IN} to generate optimized probability maps $\hat{D}$ that can reflect the confidence of benign and malignant. Specifically, with the IPO, our network fully leverages prediction information in low-level iteration layer, potentially enhancing the prediction accuracy of high-level iteration layer. In addition, the parameters in the low-level iteration layer can be optimized through the back-propagation gradients from the high-level iteration layer, as illustrated in the following formula:

$$\begin{aligned} \hat{d}_1 &= ConvBlock(SAF(f_0, \alpha_1)) \\ \hat{d}_i &= ConvBlock(SAF(f_0 \oplus \hat{d}_{i-1}, \alpha_i)), i \in [1, n] \\ \alpha_i &= \alpha_1 + k \cdot (i - 1), i \in [1, n] \end{aligned} \quad (6)$$

Here, the symbol “ $\oplus$ ” represents the concatenation operation; *ConvBlock* consists of a group of asymmetric convolutions (e.g., $Conv1 \times 5$, $Conv5 \times 1$ and $Conv1 \times 1$); n denotes the number of layers in the IPO unit, α_1 is set to 1, and k represents the step between α_{i-1} and α_i . With the lesion's feature map f_0 from the ACE-TLR part, the initial distribution D_{IN} obtained by SAF is fed into the first optimize layer of the IPO unit to produce the first optimized probability map $\hat{d}_1$. Then, $\hat{d}_1$ is contacted with f_0 , to generate optimized probability map $\hat{d}_2$ through a continuous operation based on SAF and the second optimize layer of IPO unit. The optimized probability maps $\hat{d}_{i-1}$ provides prior information for producing the next probability maps $\hat{d}_i$. In this way, we can get a group of probability maps $\hat{D}$ to aware microvascular infiltration.

D. Patient-Specific Modality Integration for Thyroid Nodules Diagnosis (PSMI-TD)

The diagnosis of thyroid nodules can benefit from a comparative analysis of gray-scale US and dynamic CEUS. However, not all extracted feature information is uniformly useful [43] or equivalent for a specific patient. In clinical diagnosis, sonographers commonly consider the dynamic informativeness of a specific frame or modality for each individual patient. This consideration is crucial for CEUS-based diagnosis with the

assistance of gray-scale US. Current models that incorporate gray-scale US as an auxiliary tool in CEUS analysis often neglect the importance of varying amounts of information provided by different features or modalities. Based on this clinical insight, we propose the implementation of PSMI to eliminate irrelevant features and dynamically integrate information at the feature-level, frame-level, and modality-level based on their respective informativeness. This approach aims to enhance diagnostic accuracy by evaluating and prioritizing different levels of informativeness. In addition, this module allows the model to fully consider and account for the impact of differences in informativeness when diagnosing thyroid nodules across different patients. Specifically, our model adaptively utilizes gray-scale US and CEUS for each individual patient, effectively evaluating the information contributed by each modality to achieve a more comprehensive diagnosis.

To evaluate the exacted feature informativeness for a specific patient, we introduce Feature Informativeness Encoding unit, as follows:

$$I^m = \psi(IE^m(H^m)) \quad (7)$$

$$H_u^m = I^m \odot H^m \quad (8)$$

where IE^m consists of a fully connection layer, $I^m \in \mathbb{R}^{d_m}$ represents a vector of feature informativeness, $H^m \in \mathbb{R}^{d_m}$ represents the feature embedding of each specific patient, d_m is the length of the vector, ψ refers to the sigmoid activation function, and “ $\odot$ ” refers to the element-wise to get useful features H_u^m filtered by the induced unit [44], $m \in \{0, 1, 2, 3\}$ represents confidence maps of gray-scale US and dynamic CEUS.

For a specific patient sample, the information provided by dynamic CEUS and gray-scale US for the diagnostic task is uncertain. Therefore, it is crucial to consider the informativeness of modalities when jointly representing the multi-modal features. Although taking the Maximum Class Probability (MCP) as the predictive value achieves good performance in the classification task, it ignores that different modalities provide varying informativeness for benign and malignant predication. Thus, we use the True Class Probability (TCP) [45] instead of MCP to measure the informativeness provided by each modality for the diagnostic task. TCP is the predicted confidence score corresponding to the true class label, which is closely related to the modality informativeness used for classification. When TCP is low, it means that the modality provides low or even incorrect information for the classification task. A fully connected process $FC(\cdot)$ for predicting benign and malignant can be described as follows:

$$P^m = FC(H_u^m), P^m = [p_0^m, p_1^m] \quad (9)$$

$$L_{entropy} = - \sum_{m=0}^3 \sum_{k=0}^1 y_k \log p_k^m \quad (10)$$

The classifier can then be trained using a cross-entropy loss function to minimize the deviation between the predicted and true values, denoted as $L_{entropy}$. where $y_{k \in \{0,1\}}$ refers to the ground-truth of benign and malignant, respectively; P^m refers to the predicted probability of the m -th modality, and M represents the number of classifier branches. Here, we utilize

the probability corresponding to the true category label as the confidence score TCP to represent the specific modality informativeness:

$$TCP = y \odot P^m = \sum_{k=0}^1 y_k \cdot p_k^m \quad (11)$$

As the true category label is unknown during the testing phase, it is not possible to directly select the corresponding TCP based on the true category label. Therefore, we introduce a $\widehat{TCP}$ prediction unit to approximate the true probabilities and train it as follows:

$$L_{conf} = \sum_{m=0}^3 \left[\left(\widehat{TCP}^m - TCP^m \right)^2 + L_{entropy,m} \right] \quad (12)$$

Lastly, we dynamically integrate the features of the different modalities depending on their informativeness, obtained by:

$$H = \left[\widehat{TCP}^0 \cdot H_u^0, \widehat{TCP}^1 \cdot H_u^1, \dots, \widehat{TCP}^m \cdot H_u^m \right] \quad (13)$$

where H represents the feature embedding after the dynamic integration of a set of dynamic CEUS features and gray-scale US features, enabling a reliable diagnosis of benign and malignant.

As shown in Fig.2, the lightweight C3D includes four 3D convolution layers. Each convolution layer is succeeded by a BN layer, with two pooling layers situated after the third convolution layer and the fourth convolution layer, respectively. Especially, we used the reweighted feature F'_{4th} generated in the TPA module to assign new weights to each frame of the video for diagnosing thyroid nodules. This approach aligns our diagnostic model more consistent with clinical diagnostic guidelines, enhancing interpretability.

In summary, we capture useful feature information within each modality by introducing gating units and use true class probabilities to evaluate the informativeness of dynamic CEUS and gray-scale US. A final classifier with cross-entropy loss is employed to obtain the final diagnosis based on dynamic integration of CEUS and gray-scale US. The classification loss function can be expressed as follows:

$$L_{cls} = L_{conf} + L_{entropy}. \quad (14)$$

E. Loss Function

In this section, we briefly describe the overall training process of our proposed model and the loss function employed during training. Initially, we pre-train the ACE-TLR part using the DICE loss function L_{Dice} to get initial lesion identification module which can output the feature f_0 . Subsequently, we proceed to train the joint segmentation and classification model using a hybrid loss function. This involves incorporating the pre-trained ACE-TLR, enabling the model to achieve automatic recognition and diagnosis of lesion regions.

With continuous probability map $\hat{D}$ obtained from MIA, $\hat{D}$ are multiplied with the input image. Then, these maps are fed into a lightweight C3D to predict the probability of benign and malignant lesions, as shown in Fig.2. To constrain the generation of $\hat{D}$, the mean square error L_{MSE} is employed.

The loss function L_{iden} of lesion areas identification can be expressed:

$$L_{MSE} = \sum_{i=1}^n \frac{1}{\Omega} \sum_{pi \in \Omega} \|g_i(pi), \hat{d}_i(pi)\|_2 \quad (15)$$

$$L_{iden} = L_{Dice} + L_{MSE} \quad (16)$$

where g_i is the label of $\hat{d}_i$, which is generated by the operation $SAF(D_{(i,j)}, \alpha_i)$, and pi denotes pixel in the image domain Ω . Assuming that $\hat{D}$ is ready to supervise the classification network, we want to ensure that the probability maps can accurately reflect the classification confidence maps and balance inconsistencies between segmentation and classification. Thus, we design a task-focused loss L_{ta} to balance confidence maps for classification and segmentation results, defined as follows:

$$L_{ta} = \frac{1}{2\eta^2} \sum_{i=1}^n \|d_i - \hat{d}_i\|_2^2 + \log \eta \quad (17)$$

η is a learnable parameter introduced to eliminate the hidden uncertainty information. Rather than simply predicting whether a thyroid nodule is benign or malignant, we have innovatively considered the specificity of each patient. Consequently, we dynamically integrate the series of probability distributions D to enhance trustworthy diagnostic outcomes.

To train the joint segmentation and classification model, we employ a hybrid loss that consists of L_{cls} , L_{iden} , and task-focused loss L_{ta} . The L_{total} is denoted as follows:

$$L_{total} = \frac{1}{2\omega_1^2} L_{cls} + \frac{1}{2\omega_2^2} L_{iden} + L_{ta} + \log \omega_1 \omega_2 \quad (18)$$

where ω_1 and ω_2 are the parameters to balance the uncertain weights and learned by the model training. We set initialized ω_1 and ω_2 are 1 in the experiments.

III. EXPERIMENT

A. Dataset and Experiment Setup

1) **Dataset:** The thyroid nodule dataset used in this study comprised 282 consecutive patients who had undergone thyroid nodule examinations at Nanjing Drum Tower Hospital. An experienced Sonographer conducted the US examinations using an iU22 scanner (Philips Healthcare, Bothell, WA) with a linear transducer L9-3 probe. The US examination involved the acquisition of both dynamic gray-scale US and CEUS images simultaneously, using the dual-image display format. Each frame of the dual-images had a resolution of 600×800 pixels. The malignant and benign status of the nodules was determined based on the diagnostic conclusions drawn from the pathological examination of percutaneous biopsies. Out of the 282 cases, 147 were identified as malignant nodules and 135 as benign nodules. A sonographer with over 10 years of experience manually annotated the pixel-level ground-truth of thyroid nodules, accurately locating the lesions using a binary mask. All data were approved by the Institutional Review Board and informed consent of the patients.

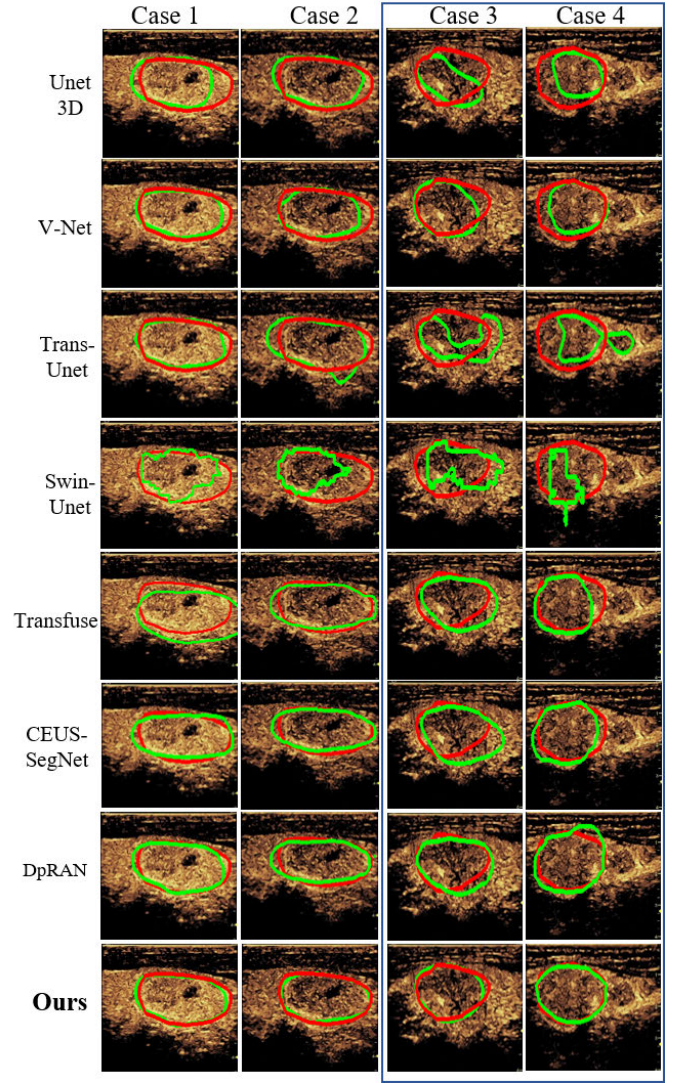


Fig. 3. Segmentation result by using different methods. Green and red contours mean the automatically segmented lesion area and ground truth. (Images have been cropped to show the results more clearly.)

2) **Implementation Details:** The model was implemented using PyTorch library and executed on a single 12 GB NVIDIA RTX 3060 GPU. The training process consisted of two stages. In the first stage, the ACE-TLR part, comprising V-Net backbone and two core modules (TPA and ACE), was pre-trained for 60 epochs using the Dice loss. We use Adam optimizer to update network parameters with the learning rate of 10^{-4} . In the second stage, the pre-trained weights were loaded to train the entire model for 100 epochs, using Adam optimizer with an initial learning rate of 10^{-4} and a weight decay rate of 10^{-3} . Data augmentation techniques, including random rotation and cropping, were applied. The resolution of the dual-image frames was resized to 256×512 , resulting in a resolution of 256×256 for both the gray-scale US and CEUS images. The parameter λ in equation (1) and θ in equation (2) are 10^{-3} and 10^{-2} , respectively. We set $\alpha = \{1, 5, 9\}$ in equation (4), which was observed to be the most effective for diagnostic tasks in Section IV-A.

3) **Evaluation Metrics:** To ensure a fair comparison, the five-fold cross validation was conducted, where one fold was

TABLE I

THE RESULTS OF DIFFERENT SEGMENTATION METHODS. THE BEST RESULTS ARE MARKED WITH BOLD TEXTS. ASTERISKS INDICATE THAT DIFFERENCE BETWEEN OUR METHOD AND THE COMPARISON METHOD IS SIGNIFICANT USING PAIRED STUDENTS' T-TEST (* : $p < 0.05$)

Method	Dice(%)	IOU(%)	Recall(%)
Unet3D	73.63±5.54*	60.56±5.83*	79.96±4.39*
V-Net	77.94±4.77*	65.20±5.09*	83.17±5.05*
Trans-unet	72.84±6.88*	59.83±6.19*	77.69±5.18*
Swin-unet	63.96±5.94*	54.43±6.91*	73.67±6.77*
Transfuse	75.68±7.63*	63.26±6.87*	86.59±6.04*
CEUS-SegNet	81.13±5.54*	70.54±4.91*	90.72±4.97
DpRAN	81.74±4.07*	72.68±4.83*	90.04±3.04*
Vnet+TPA	82.95±4.03*	73.85±4.66*	90.36±4.80*
Vnet+TPA+ACE (Our_seg)	86.02±4.34	76.14±4.23	93.52±3.63

used for the test set, another for the validation set, and the remaining folds for the training set. In the segmentation task, we utilized three indexes, namely Dice, Recall, and Intersection over Union (IOU), to evaluate the performance. Additionally, we assessed the diagnostic results using five metrics: Average Accuracy (ACC), Sensitivity (Se), Specificity (Sp), F1-score (F1), and Area Under Curves (AUC).

B. Comparison With State-of-the-Art Methods and Ablation Experiment

We first conducted separate comparisons of the segmentation and classification tasks using our model against related segmentation and classification methods that utilized CEUS and gray-scale US as input. Next, we compared the performance of our joint segmentation and classification network with multi-task or segmentation-assisted classification methods. Simultaneously, we conducted ablation experiments on all core modules of our proposed method to confirm the meaningful impact of introducing these modules in enhancing the model's performance.

1) *Comparison With Related Segmentation Methods and Ablation of Core Modules:* We conducted a quantitative and qualitative comparison of our proposed network with several related segmentation methods, including V-Net [34], Unet3D [46], TransUnet [47], Swin-unet [48], Tansfuse [49], as well as CEUS-SegNet [50] and DpRAN [51] for CEUS-based segmentation. For segmentation-assisted diagnosis, the index of Recall is important, because information in irrelevant regions can be discarded, but it will be disastrous to lose any lesion information. The experimental results in Table I demonstrate that our method achieved the highest Dice of 86.02% and recall of 93.52% compared with other methods. Fig.3 illustrates the visualization of segmentation results of different methods, displaying two benign lesions followed by two malignant lesions from left to right. Among all models, our method demonstrates the most accurate segmentation of lesion areas, as evidenced by the close resemblance between the segmented areas (green contours) and the ground truths (red contours). Notably, our model excels in challenging instances, such as small lesions with ambiguous boundaries, as indicated by the blue box in Fig.3.

Effectiveness of TPA and ACE Module: Next, to verify the effectiveness of ACE and TPA modules for lesion

TABLE II

THE RESULTS OF DIFFERENT CLASSIFICATION METHODS. THE BEST RESULTS ARE MARKED WITH BOLD TEXTS. ASTERISKS INDICATE THAT DIFFERENCE BETWEEN OUR METHOD AND THE COMPARISON METHOD IS SIGNIFICANT USING PAIRED STUDENTS' T-TEST (* : $p < 0.05$)

Method	ACC(%)	AUC(%)	Se(%)	Sp(%)	F1(%)
C3D+Mask	76.61±4.13*	88.05±3.01*	88.05±3.73*	76.92±3.17*	77.52±3.32*
R3D+Mask	77.01±3.25*	83.10±3.82*	87.17±4.08*	79.49±3.88*	83.05±3.93*
R2plus1D+Mask	78.88±2.81*	81.10±3.82*	85.02±2.49*	77.45±3.43*	81.75±2.79*
ConvLSTM+Mask	78.59±4.44*	85.95±3.73*	84.37±3.66*	76.26±4.26*	80.22±3.92*
AHAF+Mask	80.11±3.45*	90.20±3.92	90.16±3.87	78.19±4.12*	82.87±3.99*
T-AFML+Mask	79.19±4.26*	89.54±3.88	89.36±2.76	79.63±3.22*	81.04±3.05*
Baseline+Mask	79.29±2.58*	85.25±2.84*	89.83±1.80*	79.84±3.04*	83.01±2.00*
Baseline+Mask +PSMI(Our_Cla)	82.18±2.86	90.65±3.40	91.37±4.11	81.31±2.27	86.31±2.98

identification, we also compared the results of our model with and without these modules. The results are presented in Table I, revealing the following insights: 1) The inclusion of the TPA module resulted in a substantial improvement of 8.65% in IOU and 5.01% in Dice results for our model; 2) by further introducing the ACE module, the segmentation model improved by an average margin of 2.29% in terms of mean IOU and 3.07% in terms of dice, highlighting the effectiveness of the ACE module; 3) in comparison to the backbone V-Net, our model exhibited a significant improvement in Recall, with an increase of 10.35% due to the introduction of TPA and ACE modules. This improvement is particularly noteworthy for the diagnosis of thyroid nodules.

2) *Comparison With Related Classification Methods and Ablation of Core Modules:* To validate our proposed method for the diagnosis of thyroid nodules, we conducted comparisons with SOTA video classification methods such as C3D [52], R3D [53], R2plus1D [53], ConvLSTM [54] and CEUS-based classification methods of AHAF [55] and T-AFML [56]. To ensure a fair comparison, all methods utilized binary masks of lesions to aid diagnosis. In Table II, we can observe that our model generally outperforms their competitors.

Effectiveness of PSMI Module: It is important to note that the MIA module cannot be introduced when using a binary mask to assist in the diagnosis, and its validation will be addressed in next section. Here, we incorporated the PSMI module into the baseline network, resulting in a 2.89% increase in accuracy for thyroid nodule diagnosis.

3) *Comparison With Related Multi-Task Methods and Ablation of Core Modules:* In contrast to the separate approaches of lesion segmentation and diagnosis described above, our model integrates segmentation and classification through the incorporation of the MIA module and the task-focused loss function L_{ra} . In this experiment, we compared our model with related multi-task methods, including MA-MTLN [57], MTTU-Net [58], DSI-Net [59], CMSVNet [60] and Multi-task [18]. From Table III, we can see that our method demonstrates superior performance across all evaluation metrics for both segmentation and diagnosis. Compared to the second-best model, the metrics of segmentation are improved by 6.42%, 6.66% and 4.16%, respectively. The metrics of diagnosis are improved by 6.45%, 3.25%, 1.30%, 4.64% and

TABLE III

THE RESULTS OF DIFFERENT MULTITASK METHODS. THE BEST RESULTS ARE MARKED WITH BOLD TEXTS. ASTERISKS INDICATE THAT DIFFERENCE BETWEEN OUR METHOD AND THE COMPARISON METHOD IS SIGNIFICANT USING PAIRED STUDENTS' T-TEST (* : $p < 0.05$)

Method	Segmentation			Classification					
	Dice(%)	IOU(%)	Recall(%)	ACC(%)	AUC(%)	Se(%)	Sp(%)	F ₁ (%)	
MA-MTLN	76.39±4.67*	64.45±5.01*	86.57±3.78*	79.41±4.46*	85.13±3.28*	81.19±5.08*	76.57±4.17*	80.34±5.36*	
MTTU-Net	78.63±3.95*	66.54±4.51*	88.55±3.54*	81.36±3.54*	87.95±4.04*	89.08±3.11*	79.21±3.97*	84.63±3.57*	
DSI-Net	80.07±4.96*	66.94±4.80*	87.95±5.36*	80.90±5.18*	86.88±4.12*	88.50±2.80*	81.71±6.07*	83.80±4.33*	
CMSVNet	81.96±5.04*	72.36±5.31*	92.62±4.82	83.07±4.92*	91.68±4.36	92.45±1.96	82.31±3.09*	85.62±2.77	
Mul-task	79.82±3.16*	68.21±3.92*	90.63±4.01*	82.13±2.94*	91.37±3.82	91.69±4.26	79.12±3.54*	84.05±4.02*	
Seg	86.02±4.34*	76.14±4.23*	93.52±3.63	-	-	-	-	-	
Cla	-	-	-	82.18±2.86*	90.65±3.40	91.37±4.11	81.31±2.27*	86.31±2.98	
Seg+Cla	83.92±4.81*	75.04±4.73*	90.68±4.21*	82.86±2.31*	90.59±2.96	92.31±2.72	82.77±3.01*	84.90±2.89*	
Seg+Cla+MIA	86.97±4.09*	76.81±4.59*	93.92±4.16	86.87±4.05*	92.68±3.89	92.07±4.36	80.93±3.92*	84.86±4.14*	
Seg+Cla+MIA+PSMI+ L_{total} (Ours)	88.38±4.11	79.02±4.37	96.78±4.81	89.52±3.47	94.93±2.81	94.75±3.95	89.95±3.36	90.90±3.68	

5.28%, respectively. Additionally, when compared to performing the segmentation and classification tasks separately, combining the two tasks directly results in higher classification accuracy, with only a slightly lower segmentation precision.

Effectiveness of MIA and L_{total} Modules: In contrast to directly combining two tasks, our model incorporates MIA and a loss L_{total} , which resulted in a significant improvement in diagnostic accuracy by 4.01% and 2.65%, respectively. Moreover, the introduction of the MIA module and L_{total} also led to an enhancement in segmentation accuracy by 3.05% and 1.41%, respectively. Especially, the MIA module can effectively boost the performance with increased values of all metrics. The MIA module improved segmentation and classification accuracy by 3.05% and 4.01%, respectively.

C. Analysis of Our Framework

1) Effectiveness of Using Gray-Scale US for Assistance Based on Self-Adaptive Integration Mechanism: We achieve interactive fusion between CEUS and gray-scale US modalities through a self-adaptive integration mechanism without introducing any additional parameters to the model. Here, the self-adaptive integration mechanism corresponds to two key components of the overall network architecture: 1) The network adaptively fuses complementary information from different modalities based on the importance of each channel in the segmentation task. This operation corresponds to the ACE module of the network. 2) In the classification network, we measure the provided diagnostic information at the feature level, frame level, and modal level for different patients. Subsequently, we adaptively integrate this information based on the informativeness, and this operation corresponds to the PSMI module in the network.

To justify the effectiveness of the proposed self-adaptive integration mechanism, we performed ablation experiments on both thyroid nodule segmentation and classification tasks. We compared five frameworks to validate the effectiveness of the ACE and the PSMI modules, which are key components of self-adaptive integration mechanism.

1) **CEUS:** Only using CEUS modalities.

2) **CEUS+US:** Using gray-scale US for assistance without self-adaptive integration mechanism.

3) **CEUS+US+ACE:** Gray-scale US is employed to assist in the segmentation of thyroid nodules by using the ACE

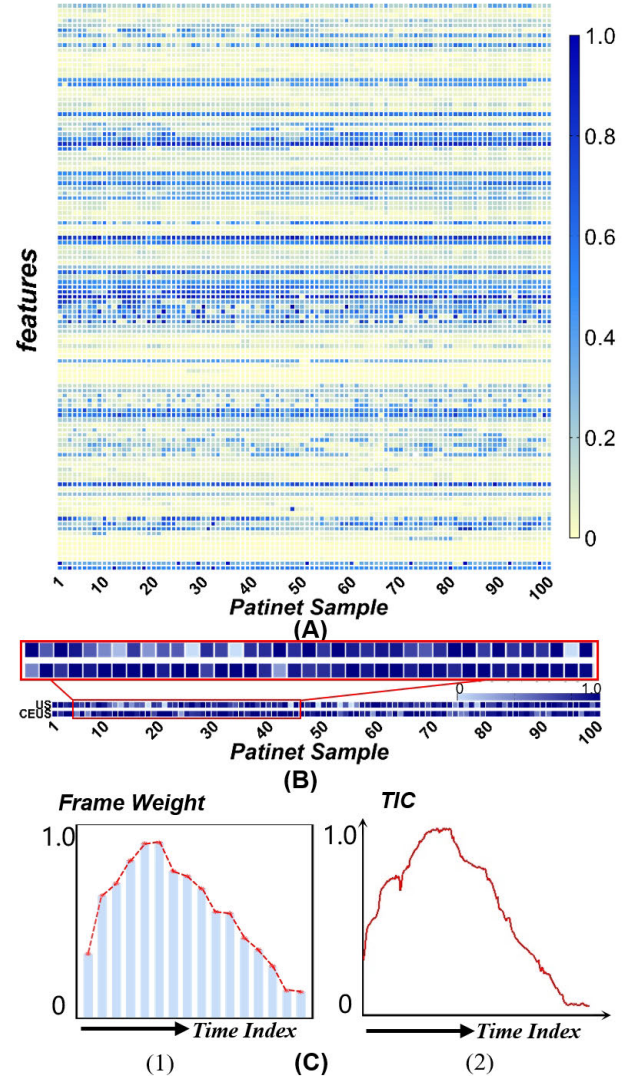


Fig. 4. (A) Feature-level informativeness; (B) Modality-level informativeness; (C) Frame-level informativeness and the clinician-labeled Time-Intensity Curve (TIC). This figure illustrates the model's capability to approximate the frame-level informativeness extracted using the TPA module to the corresponding TIC. Specifically, Figure C.(1) and Figure C.(2) display the frame-level informativeness extracted by the TPA module and the TIC labeled by clinicians, respectively.

component of the self-adaptive integration mechanism. For the subsequent task of thyroid nodule classification, the self-adaptive integration mechanism is not employed.

TABLE IV
THE RESULTS OF USING GRAY-SCALE US FOR ASSISTANCE BASED ON SELF-ADAPTIVE INTEGRATION

Modality	Segmentation			Classification				
	Dice(%)	IOU(%)	Recall(%)	ACC(%)	AUC(%)	Se(%)	Sp(%)	F ₁ (%)
CEUS	82.95±4.03	73.85±4.66	90.36±4.80	79.29±2.58	85.25±2.84	89.83±1.80	80.84±3.04	83.01±2.00
CEUS+US	84.38±3.73	74.90±3.90	91.28±3.59	80.36±2.54	88.65±3.18	90.29±3.11	80.99±2.51	84.77±2.60
CEUS+US+ACE	86.02±4.34	76.14±4.23	93.52±3.63	80.92±2.02	88.95±2.76	90.45±3.54	80.91±2.81	85.66±3.16
CEUS+US+PSMI	85.01±4.01	75.33±3.96	92.67±3.44	82.18±2.86	90.65±3.40	91.37±4.11	81.31±2.27	86.31±2.98
CEUS+US+ACE+PSMI	87.79±3.88	78.44±3.63	95.61±3.18	88.32±2.55	93.88±3.07	92.87±3.86	84.05±2.45	89.76±3.27

4) **CEUS+US+PSMI**: Gray-scale US is employed to assist in the thyroid nodule classification by using the PSMI component of the self-adaptive integration mechanism. For the subsequent task of thyroid nodule segmentation, the self-adaptive integration mechanism is not employed.

5) **CEUS+US+ACE+PSMI**: Gray-scale US is employed for both thyroid nodule segmentation and classification by using the ACE and PSMI components of the self-adaptive integration mechanism.

As depicted in Table IV, the introduction of gray-scale US enhances the performance of our segmentation branch by 1.43% on Dice metrics, and the performance of our diagnosis branch improves by 1.07% on the accuracy index. Specifically, the incorporation of the ACE component, designed for the segmentation task, leads to a notable enhancement of 1.64% in segmentation Dice. The enhanced segmentation performance directly contributes to a 0.56% increase in classification accuracy because the model can acquire more precise lesion regions. Similarly, the inclusion of the PSMI component, specifically designed for the classification task, leads to a 0.63% increase in the model's segmentation performance and a corresponding improvement in classification accuracy by 1.82%. After integrating both components of the self-adaptive integration mechanism into our model, a substantial overall improvement is observed. The segmentation Dice experiences a notable increase of 3.41%, and the classification accuracy demonstrates a remarkable enhancement of 7.96%. By comparing the results in Table IV, we can demonstrate the significance of the self-adaptive integration mechanism employed in the proposed method.

2) Visualization of Feature and Modality Informativeness:

To visualize the obtained various informativeness of specific patients, heatmaps are employed to show the informativeness at both the feature-level and modal-level, respectively. As shown in Fig. 4(A), the following observations can be made: 1) In the extracted features, many are considered as having low informativeness (tending to 0), while others are deemed invaluable (indicated by darker colors in the heatmap), significantly contributing to the classification task across all patients. 2) The informativeness of certain features for the diagnostic task undergo continuous changes between patients. Here, we can effectively filter out irrelevant information using a gating unit and assign weights to features based on their informativeness. As shown in Fig. 4(B), we can also see that the modal information provided by CEUS and gray-scale US changes continuously for a given patient. Consequently, dynamic integration of modal information becomes essential when performing multimodal joint representations. In addition,

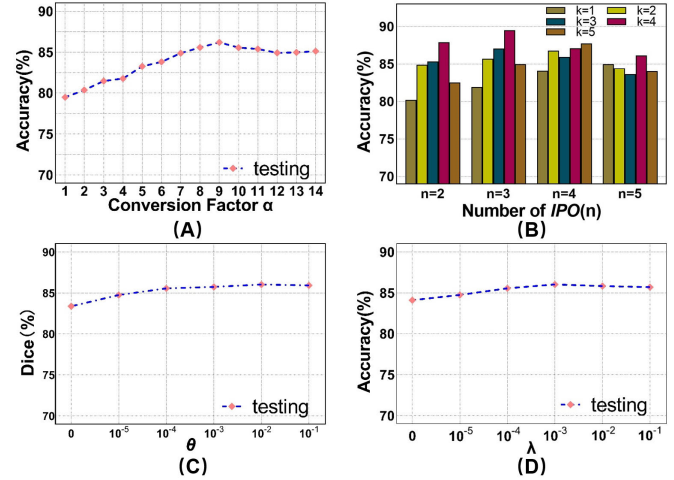


Fig. 5. The influence of hyper-parameters.

we further visualized the informativeness of different frames after the TPA module using histograms as shown in Fig. 4(C). We found that the TPA module was able to reweight each frame according to the frame-level informativeness, approximating the clinical diagnosis approach of sonographers using TIC. In summary, our model mimics the dynamic integration of different levels information by sonographers, enhancing interpretability and trustworthiness.

IV. DISCUSSION

A. Influence of Parameters

1) **Influence of Hyper-Parameters**: In the proposed method, four parameters need to be tuned: α , λ , θ , and the number of IPO denoted as n . The value of α in SAF is related to simulating microvascular infiltration. Fig.5 (A) demonstrates that as α increases, the diagnostic accuracy also increases until it reaches a plateau when α is close to 9. Furthermore, the details regarding the setting of n shown in Fig.5 (B) validate that the model performs best when $n = 3$ and the step size k of α is set to 4. Therefore, in order to balance efficiency and performance, we set the number of IPO as $n = 3$ and $\alpha = \{1, 5, 9\}$ to generate a group of confidence maps that simulate the process of microvascular infiltration.

The effect of θ (the threshold for identifying close-zero scale factors) and λ (the weights of the sparsity constraints) on the results of identifying the lesion area is illustrated in Fig. 5(C) and (D). Fig. 5(C) shows that the performance of segmentation Dice index reaches an optimal result when $\theta = 10^{-2}$. Fig. 5(D) indicates that the sparse constraint is effective in identifying

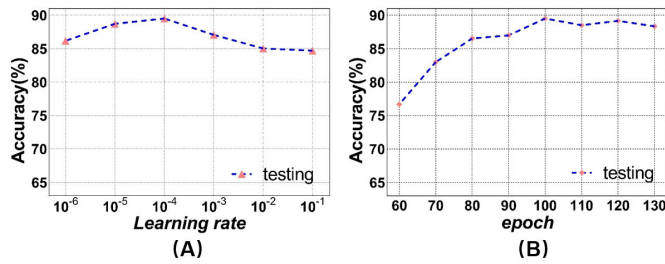


Fig. 6. The influence of training parameters.

lesion regions, and the model performs best at $\lambda = 10^{-3}$ and stabilizes afterwards.

We conducted experimental validations for these four parameters. As depicted in Fig. 5, when setting the parameters to $\alpha = 9$, $\lambda = 10^{-3}$, $\theta = 10^{-2}$, and $n=3$ for optimal performance, the model achieves its peak performance. Subsequently, the model's performance stabilizes within a certain range of parameter values, indicating its robustness to variations in those parameters.

2) Influence of Training Parameters: We employed a simple yet effective grid search algorithm to optimize our training parameters, including the learning rate and maximum epoch. Considering the computational complexity and GPU limitations, we set the batch size to 8. As depicted in Fig. 6(A), our model attains optimal performance when the learning rate is set to 0.0001. Additionally, Fig. 6(B) offers a comprehensive depiction, demonstrating that model accuracy steadily improves with an increasing number of training epochs, reaching its peak at 100 epochs.

B. Model Performance

1) Comparison With Our Previous Study: Compared to the published MICCAI work [33], we replace the Cross-task shared feature extraction (CFA) module with a new adaptive integration mechanism. Experimental results confirmed a significant enhancement in segmentation accuracy compared to the published MICCAI work. Specifically, the segmentation dice improved by 2.84%, IOU by 4.03%, and Recall by 6.38%. In comparison to the published MICCAI work, the diagnosis accuracy has slightly improved, and the Grad-CAM visualization results show that the new model's diagnosis is more reasonable, focusing more on the lesion areas. Specifically, the Grad-CAM visualization results presented in Fig. 7 demonstrate the substantial impact of textures and artifacts on the decision-making process of C3D diagnosis models. Consequently, the classification accuracy is significantly compromised. In addition, compared to previous results published in MICCAI, our Grad-CAM localization is more precise. Our model, which incorporates specific components designed to mimic the thinking process of Sonographers, effectively focuses on the critical lesion region, resulting in superior performance compared to the baseline models.

2) Model Performance Analysis: We argue that the optimal results of our proposed model are attributed to the following factors. First, the segmentation branch effectively identifies the nodule region, capturing crucial information about the size, shape and aspect ratio of the nodule. This allows the

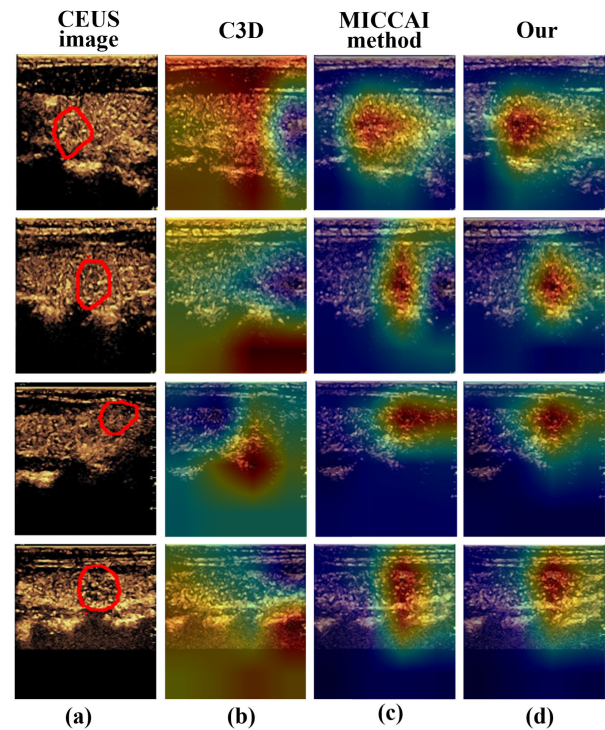


Fig. 7. Grad-CAM visualization results. (a) CEUS images with a red contour representing the lesion region outlined by a doctor; (b) Grad-CAM visualization results of C3D model; (c) Grad-CAM visualization results of the published MICCAI work; (d) Grad-CAM visualization results of our study.

classification branch to focus on the nodule region, reducing the effect of redundant information. The classification task provides the category information of lesions, and this category information is propagated to the segmentation map. This integration is leveraged to compute the loss function, effectively enhancing the segmentation performance, particularly in cases involving CEUS with uncertain boundaries. Second, we introduced the support of gray-scale US in the development of the CEUS automated diagnosis and segmentation model. As shown in Table IV and Fig. 4(B), gray-scale US was able to assist the model in identifying the lesion area and provide more enriched diagnostic information. Third, we have designed the MIA mechanism to account for the inconsistency in the size of the lesion between CEUS and gray-scale US, effectively capturing the ambiguous information caused by microvascular infiltration and incorporating clinical domain knowledge.

In order to enhance the validation of our model across diverse organ datasets, we extended our testing to include an additional breast dataset comprising 358 patients, retrieved from our previously published work [29], rather than solely relying on the existing thyroid nodules dataset. The experimental results for the additional breast dataset showed that our model also achieved a high diagnosis accuracy of 89.90% for this different organ, which validated that our model was not overfitting to the proprietary data. Additionally, to validate the effectiveness of our improved results, we conducted statistical tests using paired Student's t-test to compare the performance of our proposed method with other competing methods. The results of these tests demonstrate a statistically significant

improvement ($p < 0.05$) in both thyroid nodules segmentation and classification tasks compared to the competing methods.

C. Future Works

Researchers interested in accessing our dataset can contact us via email. The code is accessible at: <https://github.com/haozhiwen-fighting/Contrast-enhanced-Ultrasound-for-Thyroid-Nodules-Diagnosis>. There are still some limitations of our method that warrant fuller investigation in the future. Due to the lack of a publicly available dataset from multiple institutions for dynamic CEUS, our method has only been validated on our private dataset. We are currently making efforts to collaborate with more hospitals in order to establish a larger multi-center dataset in the future. It is worth noting that the proposed MIA module has only been validated in a single disease, specifically thyroid disease. There is still a challenge in addressing inconsistencies in the lesion sizes captured by dynamic CEUS and gray-scale US in diseases like breast and liver. In future studies, we aim to validate the effectiveness of our model in a broader range of diseases.

V. CONCLUSION

In this paper, we propose a deep model that emulates the thinking process of Sonographers in thyroid nodule diagnosis. Our model utilizes a set of confidence maps to reconstruct the process of lesion microvascular expansion, capturing ambiguous information resulting from microvascular infiltration and ultimately improving diagnostic accuracy. This method aims to address the inaccuracies in the diagnostic task arising from the underestimation of lesion size by gray-scale US and the general overestimation by CEUS. Additionally, our model leverages temporal information through TPA to enhance performance. The model measures informativeness and dynamically integrate it at the feature, frame, and modal levels to enhance transparency and clinical interpretability of the model's workflow. To the best of our knowledge, this is the first attempt to develop a deep model that mimics Sonographers' thinking for thyroid nodule diagnosis, considering microvascular infiltration observed on dynamic CEUS and utilizing gray-scale US for assistance.

REFERENCES

- [1] R. Liu, S. Zhou, Y. Guo, Y. Wang, and C. Chang, "Nodule localization in thyroid ultrasound images with a joint-training convolutional neural network," *J. Digit. Imag.*, vol. 33, no. 5, pp. 1266–1279, Oct. 2020.
- [2] F. Chen et al., "Deep semi-supervised ultrasound image segmentation by using a shadow aware network with boundary refinement," *IEEE Trans. Med. Imag.*, vol. 42, no. 12, pp. 3779–3793, Dec. 2023.
- [3] J. Sun et al., "TNSNet: Thyroid nodule segmentation in ultrasound imaging using soft shape supervision," *Comput. Methods Programs Biomed.*, vol. 215, Mar. 2022, Art. no. 106600.
- [4] H. Gong, J. Chen, G. Chen, H. Li, G. Li, and F. Chen, "Thyroid region prior guided attention for ultrasound segmentation of thyroid nodules," *Comput. Biol. Med.*, vol. 155, Mar. 2023, Art. no. 106389.
- [5] S.-X. Zhao, Y. Chen, K.-F. Yang, Y. Luo, B.-Y. Ma, and Y.-J. Li, "A local and global feature disentangled network: Toward classification of benign-malignant thyroid nodules from ultrasound image," *IEEE Trans. Med. Imag.*, vol. 41, no. 6, pp. 1497–1509, Jun. 2022.
- [6] J. Lu et al., "GAN-guided deformable attention network for identifying thyroid nodules in ultrasound images," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 4, pp. 1582–1590, Apr. 2022.
- [7] G. Shi et al., "Knowledge-guided synthetic medical image adversarial augmentation for ultrasonography thyroid nodule classification," *Comput. Methods Programs Biomed.*, vol. 196, Nov. 2020, Art. no. 105611.
- [8] X. Wu et al., "CacheTrack-YOLO: Real-time detection and tracking for thyroid nodules and surrounding tissues in ultrasound videos," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 10, pp. 3812–3823, Oct. 2021.
- [9] Z. Liu and C. Li, "Correlation of lymph node metastasis with contrast-enhanced ultrasound features, microvessel density and microvessel area in patients with papillary thyroid carcinoma," *Clin. Hemorheology Microcirculation*, vol. 82, no. 4, pp. 361–370, Dec. 2022.
- [10] E. Quaia, "Assessment of tissue perfusion by contrast-enhanced ultrasound," *Eur. Radiol.*, vol. 21, no. 3, pp. 604–615, Mar. 2011.
- [11] H. Zhao et al., "Diagnostic performance of thyroid imaging reporting and data system (TI-RADS) alone and in combination with contrast-enhanced ultrasonography for the characterization of thyroid nodules," *Clin. Hemorheology Microcirculation*, vol. 72, no. 1, pp. 95–106, Jul. 2019.
- [12] P. Sidhu et al., "The EFSUMB guidelines and recommendations for the clinical practice of contrast-enhanced ultrasound (CEUS) in non-hepatic applications: Update 2017 (long version)," *Ultraschall der Medizin Eur. J. Ultrasound*, vol. 39, no. 2, pp. e2–e44, Apr. 2018.
- [13] V. Cantisani et al., "Growing indications for CEUS: The kidney, testis, lymph nodes, thyroid, prostate, and small bowel," *Eur. J. Radiol.*, vol. 84, no. 9, pp. 1675–1684, Sep. 2015.
- [14] M. Brandenstein et al., "Multiparametric sonographic imaging of thyroid lesions: Chances of B-mode, elastography and CEUS in relation to pre-operative histopathology," *Cancers*, vol. 14, no. 19, p. 4745, Sep. 2022.
- [15] L. Qin, H. Yin, H. Zhuang, Y. Luo, P. Liu, and D. C. Liu, "Classification for rectal CEUS images based on combining features by transfer learning," in *Proc. 3rd Int. Symp. Image Comput. Digit. Med.*, Aug. 2019, pp. 187–191.
- [16] Z. Yang, X. Gong, Y. Guo, and W. Liu, "A temporal sequence dual-branch network for classifying hybrid ultrasound data of breast cancer," *IEEE Access*, vol. 8, pp. 82688–82699, 2020.
- [17] F. Chen, H. Han, P. Wan, H. Liao, and D. Zhang, "Joint segmentation and differential diagnosis of thyroid nodule in contrast-enhanced ultrasound images," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 9, pp. 2722–2732, Apr. 2023.
- [18] H. Gan et al., "A multi-task learning framework for carotid plaque segmentation and classification from ultrasound images," 2023, *arXiv:2307.00583*.
- [19] Y. Xie, J. Zhang, Y. Xia, and C. Shen, "A mutual bootstrapping model for automated skin lesion segmentation and classification," *IEEE Trans. Med. Imag.*, vol. 39, no. 7, pp. 2482–2493, Jul. 2020.
- [20] Y. Xie, H. Liao, D. Zhang, and F. Chen, "Uncertainty-aware cascade network for ultrasound image segmentation with ambiguous boundary," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Singapore: Springer, Sep. 2022, pp. 268–278.
- [21] H. Qian, Z. Shen, D. Zhou, and Y. Huang, "Intratumoral and peritumoral radiomics model based on abdominal ultrasound for predicting ki-67 expression in patients with hepatocellular cancer," *Frontiers Oncol.*, vol. 13, Aug. 2023, Art. no. 1209111.
- [22] L. Huang, S. Ruan, Y. Xing, and M. Feng, "A review of uncertainty quantification in medical image analysis: Probabilistic and non-probabilistic methods," 2023, *arXiv:2310.06873*.
- [23] Y. Zhang, X. Zhang, J. Li, Q. Cai, Z. Qiao, and Y. K. Luo, "Contrast-enhanced ultrasound: A valuable modality for extracapsular extension assessment in papillary thyroid cancer," *Eur. Radiol.*, vol. 31, no. 7, pp. 4568–4575, Jul. 2021.
- [24] Y.-X. Jiang, H. Liu, J.-B. Liu, Q.-L. Zhu, Q. Sun, and X.-Y. Chang, "Breast tumor size assessment: Comparison of conventional ultrasound and contrast-enhanced ultrasound," *Ultrasound Med. Biol.*, vol. 33, no. 12, pp. 1873–1881, Dec. 2007.
- [25] H.-T. Hu et al., "Ultrasound-based radiomics score: A potential biomarker for the prediction of microvascular invasion in hepatocellular carcinoma," *Eur. Radiol.*, vol. 29, no. 6, pp. 2890–2901, Jun. 2019.
- [26] M. Radzina et al., "Performance of contrast-enhanced ultrasound in thyroid nodules: Review of current state and future perspectives," *Cancers*, vol. 13, no. 21, p. 5469, 2021.
- [27] P. Wan, F. Chen, C. Liu, W. Kong, and D. Zhang, "Hierarchical temporal attention network for thyroid nodule recognition using dynamic CEUS imaging," *IEEE Trans. Med. Imag.*, vol. 40, no. 6, pp. 1646–1660, Jun. 2021.
- [28] C. Chen, Y. Wang, J. Niu, X. Liu, Q. Li, and X. Gong, "Domain knowledge powered deep learning for breast cancer diagnosis based on contrast-enhanced ultrasound videos," *IEEE Trans. Med. Imag.*, vol. 40, no. 9, pp. 2439–2451, Sep. 2021.

- [29] F. Chen et al., "Immunohistochemical index prediction of breast cancer by using hybrid ultrasound data," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 4, pp. 1401–1412, Apr. 2023.
- [30] S.-H. Shao, C.-X. Li, M.-H. Yao, G. Li, X. Li, and R. Wu, "Incorporation of contrast-enhanced ultrasound in the differential diagnosis for breast lesions with inconsistent results on mammography and conventional ultrasound," *Clin. Hemorheology Microcirculation*, vol. 74, no. 4, pp. 463–473, Jun. 2020.
- [31] J.-W. Feng et al., "A nomogram based on clinical and ultrasound characteristics to predict lymph node metastasis of papillary thyroid carcinoma," *Frontiers Endocrinology*, vol. 12, Apr. 2021, Art. no. 666315.
- [32] Z. Han, F. Yang, J. Huang, C. Zhang, and J. Yao, "Multimodal dynamics: Dynamical fusion for trustworthy multimodal classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 20675–20685.
- [33] H. Han et al., "Thyroid nodule diagnosis in dynamic contrast-enhanced ultrasound via microvessel infiltration awareness," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2023, pp. 169–179.
- [34] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
- [35] Y. Wang et al., "Key-frame guided network for thyroid recognition using ultrasound videos," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2022, pp. 238–247.
- [36] Y. Wang, F. Sun, W. Huang, F. He, and D. Tao, "Channel exchanging networks for multimodal and multitask dense image prediction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 5, pp. 5481–5496, May 2023.
- [37] H. Wang, Y. Zhu, B. Green, H. Adam, A. Yuille, and L.-C. Chen, "Axial-DeepLab: Stand-alone axial-attention for panoptic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 108–126.
- [38] Y. Jiang et al., "APAUNet: Axis projection attention Unet for small target in 3D medical segmentation," in *Proc. Asian Conf. Comput. Vis.*, 2022, pp. 283–298.
- [39] S.-X. Zhang, X. Zhu, L. Chen, J.-B. Hou, and X.-C. Yin, "Arbitrary shape text detection via segmentation with probability maps," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 2736–2750, Mar. 2023.
- [40] W. Gómez-Flores and J. Hernández-López, "Assessment of the invariance and discriminant power of morphological features under geometric transformations for breast tumor classification," *Comput. Methods Programs Biomed.*, vol. 185, Mar. 2020, Art. no. 105173.
- [41] A. S. Mattingly, J. E. Noel, and L. A. Orloff, "A closer look at 'taller-than-wide' thyroid nodules: Examining dimension ratio to predict malignancy," *Otolaryngology-Head Neck Surgery*, vol. 167, no. 2, pp. 236–241, Aug. 2022.
- [42] Y. Li et al., "Pixel-anchor: A fast oriented scene text detector with combined networks," 2018, *arXiv:1811.07432*.
- [43] B.-B. Jia and M.-L. Zhang, "Multi-dimensional classification via selective feature augmentation," *Mach. Intell. Res.*, vol. 19, no. 1, pp. 38–51, Feb. 2022.
- [44] J. Arevalo, T. Solorio, M. Montes-y-Gómez, and F. A. González, "Gated multimodal units for information fusion," 2017, *arXiv:1702.01992*.
- [45] C. Corbière, "Addressing failure prediction by learning model confidence," in *Proc. Adv. Neural Inf. Process.*, vol. 32, 2019, pp. 1–12.
- [46] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.
- [47] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.
- [48] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 205–218.
- [49] Y. Zhang, H. Liu, and Q. Hu, "TransFuse: Fusing transformers and CNNs for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2021, pp. 14–24.
- [50] Z. Meng, Y. Zhu, X. Fan, J. Tian, F. Nie, and K. Wang, "CEUSegNet: A cross-modality lesion segmentation network for contrast-enhanced ultrasound," in *Proc. IEEE 19th Int. Symp. Biomed. Imag. (ISBI)*, Mar. 2022, pp. 1–5.
- [51] P. Wan, F. Chen, and D. Zhang, "Dynamic perfusion representation and aggregation network for nodule segmentation using contrast-enhanced us," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 7, pp. 3431–3442, Apr. 2023.
- [52] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4489–4497.
- [53] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, and M. Paluri, "A closer look at spatiotemporal convolutions for action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6450–6459.
- [54] R. Mutegeki and D. S. Han, "A CNN-LSTM approach to human activity recognition," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIC)*, Feb. 2020, pp. 362–366.
- [55] X. Gong, S. Yuan, Y. Xiang, L. Fan, and H. Zhou, "Domain knowledge-guided adversarial adaptive fusion of hybrid breast ultrasound data," *Comput. Biol. Med.*, vol. 164, Sep. 2023, Art. no. 107256.
- [56] P. Wan et al., "Transport-based anatomical-functional metric learning for liver tumor recognition using dual-view dynamic CEUS imaging," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 3, pp. 1012–1023, Mar. 2023.
- [57] Y. Zhang et al., "3D multi-attention guided multi-task learning network for automatic gastric tumor segmentation and lymph node classification," *IEEE Trans. Med. Imag.*, vol. 40, no. 6, pp. 1618–1631, Jun. 2021.
- [58] J. Cheng, J. Liu, H. Kuang, and J. Wang, "A fully automated multimodal MRI-based multi-task learning for glioma segmentation and IDH genotyping," *IEEE Trans. Med. Imag.*, vol. 41, no. 6, pp. 1520–1532, Jun. 2022.
- [59] M. Zhu, Z. Chen, and Y. Yuan, "DSI-net: Deep synergistic interaction network for joint classification and segmentation with endoscope images," *IEEE Trans. Med. Imag.*, vol. 40, no. 12, pp. 3315–3325, Dec. 2021.
- [60] Y. Zhou et al., "Multi-task learning for segmentation and classification of tumors in 3D automated breast ultrasound images," *Med. Image Anal.*, vol. 70, May 2021, Art. no. 101918.